Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case.

Tableau is a leading **Data Visualization** and **Business Intelligence** tool used to transform raw data into interactive, easy-to-understand dashboards.

Why it is Preferred Over Spreadsheets

- **Speed & Scale:** It handles millions of rows of data without lagging, unlike traditional spreadsheets which often crash with "Big Data."
- **Automation:** It connects to live data sources, meaning dashboards update automatically as new information arrives—no manual re-entry needed.
- **Visual Discovery:** It uses a "drag-and-drop" interface designed specifically for visual storytelling and "drill-down" analysis rather than just cell-based calculations.

Industry Usage

Industry	**Use Case**
Retail & E-commerce	**Sales Performance:** Tracking real-time sales across different regions to identify high-performing products and optimize inventory.
Healthcare	**Patient Monitoring:** Visualizing hospital bed occupancy and ER wait times to improve staff allocation and patient flow.

Q2. After connecting to the Sample – Superstore dataset: a) List the main types of fields available (examples: time-based, geographical, numerical). b) Identify any four Dimensions and any four Measures from the dataset.

A) main types of data fields are –

1.category

2.city

3.region

4.order id

5.order date

5.product name

6.product id

7.discout

8.profit

9.sales

b) Four dimensions are –

category , region ,order id , product name

Four measures are-

Profit, sales, discount, quantity

Q3. Create a Bar Chart showing Total Sales by Category. Explain: Which field goes to Rows Which field goes to Columns Which aggregation is applied to Sales

To create a bar chart in Tableau showing **Total Sales by Category**, follow these steps:

1. Field Placements

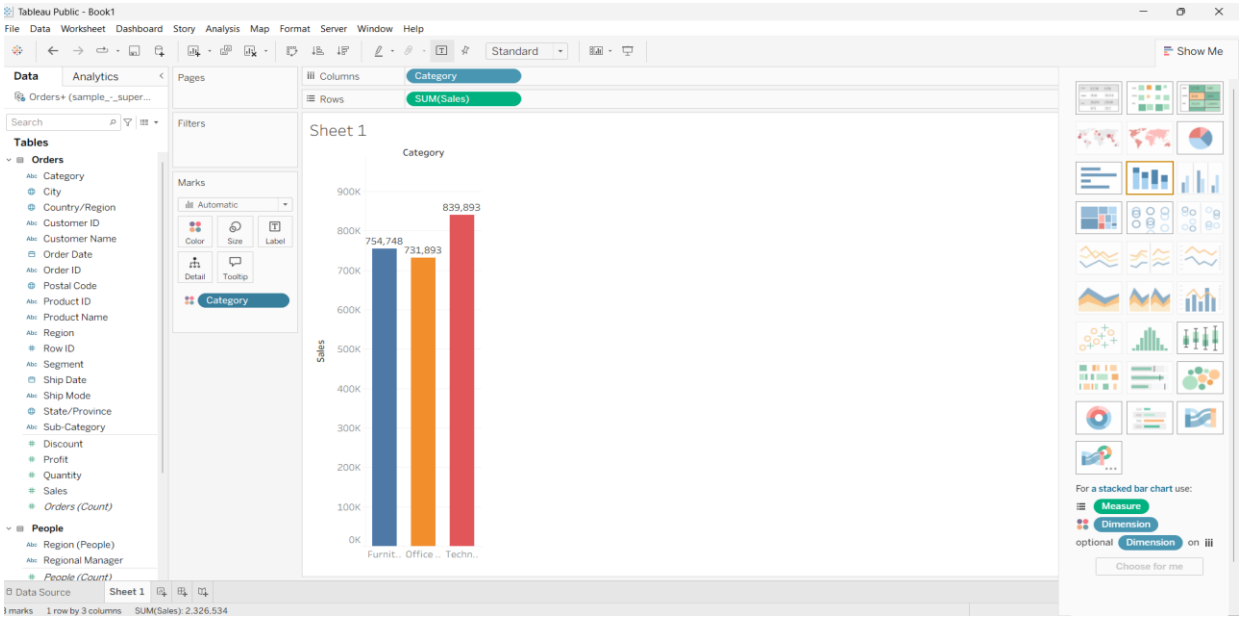
- **Columns Shelf:** Place the **Category** field here. (This creates the horizontal labels for each bar).
- **Rows Shelf:** Place the **Sales** field here. (This determines the height of the bars).

*Note: If you swap these—placing Category on Rows and Sales on Columns—you will create a **Horizontal Bar Chart**.*

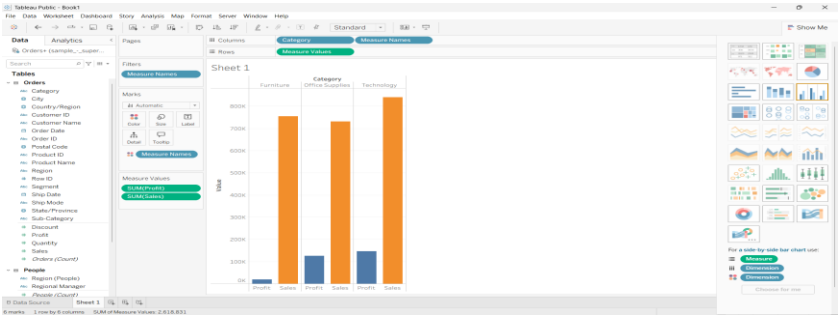
2. Aggregation Applied

- **Aggregation: SUM**
- **Explanation:** By default, when you drag a measure like "Sales" into the view, Tableau applies the SUM aggregation. This adds up every individual transaction amount within each category to show the "Total Sales."

0



Q4. What insight can be gained by comparing Sales and Profit together? Then explain how you would create a chart to analyze Profit by Sub-Category.



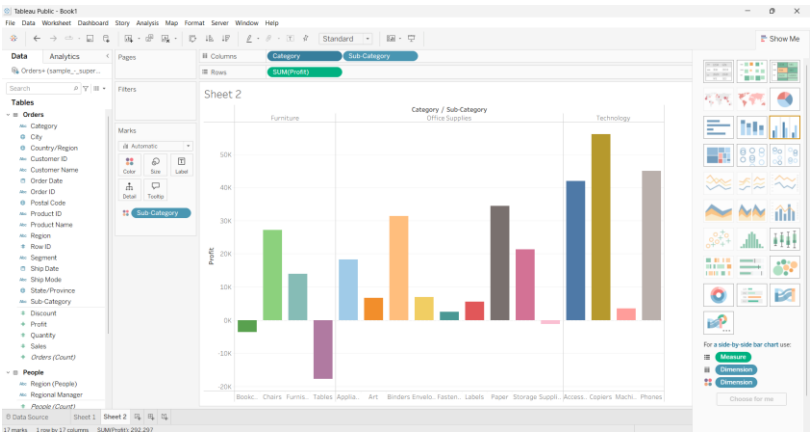
Category	Sales Insight	Profit Insight
Technology	Highest overall revenue.	Most profitable category.
Office Supplies	Lowest sales of the three.	Better margins than Furniture.
Furniture	High demand/sales.	Critically low profit margins.

Then explain how you would create a chart to analyze Profit by Sub-Category.

To create a chart for **Profit by Sub-Category** in short:

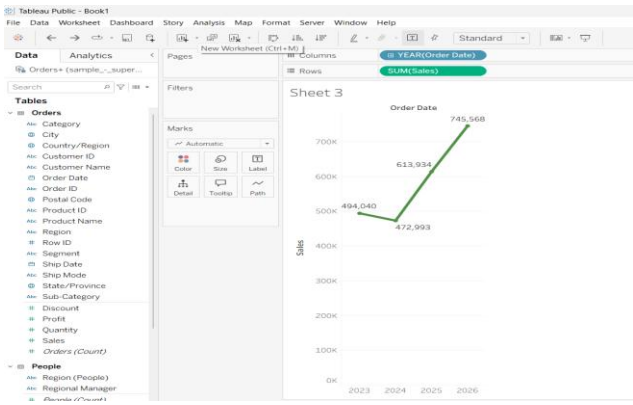
1. Field Placements

- **Columns Shelf:** Drag **Profit** here (this creates the horizontal axis).
- **Rows Shelf:** Drag **Sub-Category** here (this lists specific items like Tables, Phones, etc.).
- **Color (Marks Card):** Drag **Profit** onto Color to instantly see which items are losing money (usually shown in red/orange).



Q5. Create a Line Chart to show Sales Trend over Order Date. Which date level would you choose (Year / Quarter / Month)? Why are line charts suitable for time-based analysis?

Ans: I will choose year level



Line charts are preferred for time-based analysis because they focus on **continuity** and **change**.

Key Reasons

- **Trend Visibility:** Connecting dots with a line makes it easy to see the "slope"—instantly showing if numbers are rising, falling, or flat.
- **Flow of Time:** They follow a natural left-to-right sequence, mirroring how we experience time.
- **Pattern Recognition:** They highlight **seasonality** (e.g., holiday spikes) and **outliers** (sudden drops) much more clearly than bars.
- **Cleanliness:** They can handle hundreds of data points (like daily sales) without becoming cluttered or "noisy."

Q6. Apply a Basic Filter to display data only for: Region = “West” Category = “Technology” Explain the steps and mention which shelf is used.

The Steps

1. **For Region:** Drag the **Region** field from the Data pane and drop it onto the **Filters Shelf**. A dialog box will appear; check the box for **"West"** and click OK.
2. **For Category:** Drag the **Category** field from the Data pane and drop it onto the **Filters Shelf**. In the dialog box, check the box for **"Technology"** and click OK.

Which Shelf is Used?

- **The Filters Shelf:** Located directly above the Marks card. It is used to include or exclude specific data points from your entire view.

Q7. Explain the business importance of geographical analysis. Then describe how you would create a Map showing Sales by State using Sample – Superstore.

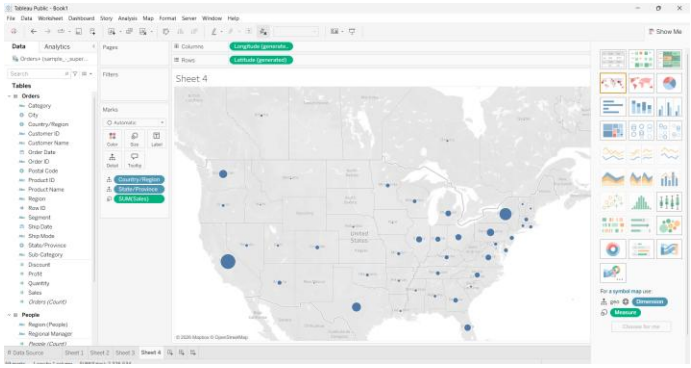
Geographical analysis allows businesses to move beyond "how much" they are selling to **"where"** the growth or decline is happening.

- **Resource Allocation:** Identify high-performing regions to focus marketing budgets or open new branches.
- **Logistics Optimization:** Pinpoint clusters of high demand to streamline supply chains and reduce shipping costs.
- **Market Penetration:** Spot "cold spots" (low sales areas) where the brand is underrepresented compared to competitors.
- **Regional Trends:** Understand how local factors (weather, culture, or regional economy) affect buying habits.

Steps to Create a Sales Map

To create a map in Tableau using the **Sample – Superstore** dataset:

1. **Generate the Map:** Double-click the **State/Province** field. Tableau will automatically place *Latitude* on Rows and *Longitude* on Columns, creating a map.
2. **Add Data:** Drag the **Sales** field onto the **Color** square in the **Marks Card**.
3. **Refine the View:** Change the Mark type from *Automatic* to **Map** (Filled Map) to color each state based on its total sales volume.



Q8. Worksheet vs. Dashboard

The Difference

- **Worksheet:** A single workspace where you create one specific visualization (e.g., one bar chart or one map). It is the building block of your analysis.
- **Dashboard:** A collection of several worksheets displayed together in a single view. It allows you to compare different data points simultaneously.

Why Dashboards are Better for Decision-Makers

- **The Big Picture:** Leaders don't have time to toggle through 10 individual charts. A dashboard provides an "at-a-glance" summary of the entire business.

- **Interactivity:** Dashboards allow users to click a "Region" in one chart and see all other charts on the page filter automatically.
- **Context:** Seeing Sales alongside Profit and Growth helps leaders understand *why* numbers are moving, not just *what* the numbers are.

Q9. Tableau Prep & Data Cleaning

What is Tableau Prep?

Tableau Prep is a visual data preparation tool designed to **clean, shape, and combine** data before it is used for analysis. In the real world, data is rarely "ready-to-use"—it's often messy, inconsistent, or formatted incorrectly. Prep automates the tedious work of fixing these issues.

How to Clean Data (The Steps)

1. **Remove Null Values:** Click on the field containing nulls. Select the "Null" bar in the profile pane, right-click, and choose **Exclude**. Alternatively, use a "Filter" to keep only non-null values.
2. **Remove Duplicate Records:** Use the **Aggregate** step. Group by all your header fields; this collapses multiple identical rows into a single unique row.
3. **Change Data Types:** Look at the data type icon (e.g., # for numbers, Abc for text) above the column header. Click the icon and select the correct type (e.g., change a "String" Date to a "Date" type).

Q10. Retail Manager's Case Study

a) Recommended Charts

- **Highest Profit by Category:** Bar Chart (Sorted descending).
- **Poorly Performing Region:** Filled Map or Highlight Table (Using color to show low-profit areas).
- **Sales Over Time:** Line Chart (With Month/Year on the horizontal axis).

b) Key Insight

Example Insight: "While the Furniture category has the second-highest sales, it actually has the lowest profit margin, particularly in the Central region."

c) Impact on Business Decision-Making

This helps managers decide where to **cut costs** or **increase prices**. Instead of just pushing for more sales, they can pivot their strategy to focus on the most profitable products, ensuring the business grows sustainably rather than just getting "bigger" without making money.